

A-level

ENVIRONMENTAL SCIENCE

Paper 2 - Generated practice paper Set 18

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

- The number in brackets after each marking point is the mark it earns; the note above the points says how many are needed.
- Where a question asks for a fixed number of points (eg *two reasons*), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	<i>Any 2 from the points below - 1 mark each, maximum 2.</i> • Potato bean (N America): high protein content (1) • Morama bean (southern Africa): protein content similar to soya, drought resistant (1) • Yeheb tree (Somalia): edible nuts, drought resistant, grows on poor soils (1) • Perennial maize (Mexico) / Kernza: no need to buy seed each year, less ploughing so less soil erosion (1) • American bison: graze grassland too poor for cattle (1) • Common eland: browse and graze so can be kept with cattle without severe competition (1)	2
01.2	<i>1 mark for each point.</i> • Indigenous species are better adapted to local conditions such as climate, pests / diseases and soil (1) • So they have higher survival rates / productivity is not reduced, as it is when introduced species are kept in areas to which they are not adapted (1)	2
01.3	<i>1 mark.</i> • They grow in soils with a high salt content, so their genes could be used in breeding programmes to increase the salt tolerance of crops grown on soils salinised by irrigation (1)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 11; AQA spec 3.1.2.1.1.	

Question	Marking guidance	Marks
02.1	<i>1 mark for each point.</i> • MEDCs: market demand and higher prices for the flavour of carnivorous fish (salmon, trout) justify high inputs (fishmeal, cooler climates suit them) (1) • LEDCs: herbivores / omnivores feed on natural vegetation, phytoplankton and wastes so need low inputs; warm climates suit tilapia (1) • Herbivores are more energy-efficient (lower trophic level) so more food per unit area (1)	3
02.2	<i>1 mark for each point.</i> • Some genetically female (XX) fish are treated with testosterone so they become functional males producing only X-carrying sperm (1) • Crossed with normal females (XX) all offspring are XX females (1) • Female trout meat has a better flavour; in tilapia the reverse (all males) because males grow larger / more efficiently (1)	3
02.3	<i>1 mark for each point.</i> • Salmon need high oxygen: water sprays / weirs aerate; wastes removed to prevent deoxygenation (1) • Constant circular flow keeps fish swimming the same way - fewer collisions / injuries, more muscular fish, higher stocking density (1) • Artificial lighting delays maturation (salmon stop growing at maturity) so larger fish; controls smoltification (1) • Cool sites (Scotland, Norway) because warm water holds less oxygen although warmth increases growth (1)	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 352-355; AQA spec 3.5.2.3.	
03.1	<i>1 mark for each point.</i> • Less dead organic matter / leaf litter so less humus and slower soil formation; fewer nutrients returned (1) • Loss of canopy and litter exposes soil to rain splash and wind (1) • Decomposing roots no longer bind soil so erosion, gullyng and landslides increase (especially on slopes) (1) • Reduced organic matter and biota lower infiltration and water retention (1)	4
03.2	<i>1 mark for each point.</i> • Reduced interception and transpiration so less water returned to the atmosphere and less rainfall downwind (1) • Increased runoff: more flooding after rain, lower flows in dry periods; sedimentation of rivers (1) • Increased albedo of cleared land (reflects more sunlight) (1) • Reduced carbon sequestration and loss of the carbon reservoir (CO₂ released by burning / decay) contributing to climate change (1)	4
03.3	<i>Any 2 from the points below - 1 mark each, maximum 2.</i> • Fragmentation of remaining forest into isolated patches with non-viable populations / edge effects (1) • Loss of inter-species relationships (pollinators, seed dispersers) causing further declines (1) • Loss of genetic resources / potential medicines before they are discovered (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 368-369; AQA spec 3.5.3.3.	

Question	Marking guidance	Marks
04.1	<i>1 mark for each point.</i> • Cycling of materials (like biogeochemical cycles) - recycling, re-use (1) • Energy from renewable sources (1) • Human activities should support ecosystems / conserve biodiversity (1) • Separation of technical and biological materials; connected systems where the waste of one process feeds another (industrial symbiosis) (1) • Diverse systems are more resistant to change (1) • Design of products for extended use and end-of-life re-use; optimum rather than maximum production; technologies that improve efficiency and enable dismantling (1)	6
04.2	<i>1 mark for each point.</i> • Natural systems over-produce so the surplus supports the processes the system relies on: plant products feed pollinators, seed dispersers, decomposers, mycorrhizae (1) • Agroecosystems aim for maximum harvested yield, removing the surplus that would maintain soil biota, pollinators and pest predators, so they need artificial inputs (1)	2
04.3	<i>1 mark for each point.</i> • Costs of environmental measures could put companies in one country at a disadvantage against competitors in countries without them (1) • A common approach gives a level playing field, coordinated monitoring and support for less successful countries (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 374-375; AQA spec 3.6.4.1.	
05.1	<i>1 mark for each point.</i> • Single genes can be transferred from any species, not just closely related ones that produce fertile hybrids (1) • Desired characteristics introduced without linked unwanted ones; faster (1)	2
05.2	<i>1 mark for each point.</i> • Bt: reduces insecticide spraying so fewer non-target deaths, lower cost / energy (1) • Roundup Ready: weeds controlled by spraying glyphosate without harming the crop, so higher yields with less cultivation / tillage (1) • Golden rice: provides vitamin A in the staple diet, reducing about 500 000 cases of blindness a year in India / sub-Saharan Africa (1)	3
05.3	<i>1 mark for each point.</i> • Gene transfer by pollen to conventional / organic crops or wild relatives (eg Bt gene killing insects feeding on wild plants) (1) • Antibiotic-resistance marker genes could transfer to pathogenic bacteria; possible allergies; gene transfer to gut bacteria (1) • Detection in the human food chain from crops approved only for animal feed (1) • Economic: seeds must be bought each year from patent-holding companies, raising costs and reducing local crop diversity and seed saving in LEDCs; herbicide-resistant crops may increase herbicide use (1)	4
05.4	<i>1 mark.</i> • Public / political concern that risks are not fully assessed - more evidence needed (precautionary principle) (1)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 323-324; AQA spec 3.5.1.2.	

Question	Marking guidance	Marks
06.1	<i>1 mark for each point.</i> - Natural cycles use a small number of abundant elements (C, H, O, N, P) built into monomers and polymers (carbohydrates, proteins); wastes of one process are raw materials for another so nothing is depleted or accumulates (1) - Natural wastes are biodegradable / non-toxic and are broken down by decomposers into raw materials (1) - Human systems are linear: extract - process - manufacture - use - discard; minerals are dispersed after use or mixed so recovery is difficult (1) - Human wastes are often persistent and toxic (organochlorines, PCBs, heavy metals in electronics) and build up; fossil fuel use is inherently linear and unsustainable (1)	4
06.2	<i>1 mark for each point.</i> - Metals are dispersed in small amounts within complex mixed products so they cannot easily be separated / recovered at end of life (1) - They leach from landfill or are released by incineration; persistent, bioaccumulating (1)	2
06.3	<i>1 mark for each point.</i> - Design for durability, repair and re-use (replaceable parts, no built-in batteries, spare parts and support, no planned obsolescence) (1) - Easy identification of materials (coding) and separation of components so they can be recycled (1) - Separation of technical materials (metals, plastics) from biological materials (which can be composted / digested) (1) - Use of recycled and recyclable / biodegradable materials; reduced packaging and material use (Cradle to Cradle) (1)	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 373-374; AQA spec 3.6.3.	
07.1	<i>1 mark for each point.</i> - Anchovy r-selected; blue whale k-selected (1) - r-selected: mature quickly, produce many young, disperse widely, short lives; adapted to respond rapidly to high death rates / unstable conditions (mice, locusts, greenfly, rabbits) (1) - k-selected: mature late, few young with parental care, long-lived; populations stable near carrying capacity (whales, elephants, rhinos) (1)	3
07.2	<i>1 mark for each point.</i> - Low reproductive rate means losses from hunting / habitat change cannot be replaced quickly (1) - An increase in death rate above the low birth rate causes a population crash (1) - Recovery takes decades (eg blue whale still far below pre-whaling numbers after the 1986 moratorium); r-selected species rebound within a few years once exploitation stops (1)	3
07.3	<i>1 mark.</i> New population = starting population + births + immigrants - deaths - emigrants (1)	1

Question	Marking guidance	Marks
07.4	<i>1 mark for each point.</i> • The greatest rate of exploitation (harvest) possible without causing long-term population decline (1) • Equals the surplus production: highest when the population is well below carrying capacity where growth rate is fastest (middle of the sigmoid curve) (1) • At carrying capacity net growth is zero so any harvest reduces the population; harvesting above MSY causes decline (1)	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 79-80; AQA spec 3.1.3.4.	
08.1	<i>1 mark for each point.</i> • A community maintained at a stage before the climax by regular human activity (grazing, mowing, burning, coppicing) that deflects succession (1) • Chalk grassland was produced by centuries of sheep grazing; without it succession to scrub and woodland resumes (1)	2
08.2	<i>1 mark for each point.</i> • Grazing removes taller / dominant grasses and prevents scrub establishing (5% cover) so small plants are not shaded out (1) • Trampling creates bare ground for wildflower seed germination and warm patches for invertebrates / reptiles (1) • Dung creates small nutrient-rich patches supporting different plants and invertebrates; sheep eat short grasses and some bushes (1) • Mowing removes everything at once including plants grazers avoid, and removes flowers / seeds before setting; burning removes vegetation but also kills invertebrates and allows scrub regrowth; abandoned plot dominated by scrub (55%) so few species (1)	4
08.3	<i>1 mark for each point.</i> • Sheep eat short grass (some breeds eat scrub); ponies avoid flowers but eat bracken; cattle remove long rough grass (1) • Choice depends on the vegetation to be controlled and the plants to be protected (1)	2
08.4	<i>1 mark for each point.</i> • Removes mature heather bushes and young trees that grazers will not eat, creating open unshaded ground (1) • Heat stimulates dormant heather seeds to germinate so young heather re-establishes (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 47-49; AQA spec 3.1.3.3.	
09.1	<i>Correct answer only.</i> • B (1)	1
09.2	<i>Any 4 from the points below - 1 mark each, maximum 4.</i> • Deforestation (1) • Ploughing of grassland / expansion of farmland (1) • Reservoir creation / flooding (1) • Mineral extraction, especially open-cast mining (1) • Urban expansion (1)	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 20-26; AQA spec 3.1.2.2.	

Question	Marking guidance	Marks
10.1	<i>Any 4 from the points below - 1 mark each, maximum 4.</i> • Zooxanthellae are photosynthetic algae living inside the polyp cells (1) • The algae absorb CO₂ and excretory wastes from the polyp for photosynthesis and growth (efficient nutrient recycling) (1) • They supply the polyp with oxygen and glucose - over 95% of its food energy (1) • In return the algae gain a protected habitat and nutrients they could not absorb from the nutrient-deficient water (1) • Polyps also feed on plankton captured by cilia and nematocysts (1)	4
10.2	<i>1 mark for each point.</i> • Corals deposit a skeleton of calcium carbonate (calcite) (1) • Each carbonate molecule requires the absorption of a molecule of carbon dioxide, so reefs are a carbon sink (1)	2
10.3	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) • Importance: cover 1% of ocean area but support over a quarter of marine species; fisheries (fish, lobsters, conch); medicines (AZT, Ara-C, dolastatin, secosteroids, bryostatin, trabectedin, bone grafts); climate control as carbon sink; erosion protection of low-lying islands; tourism income • Threats: physical damage (anchors, divers, litter, coral extraction), collection of ornaments (CITES Appendix II), sedimentation from rivers and coastal development, pollution (oil, nutrients causing algal shading, sewage, pesticides, sunscreen), ocean acidification, bleaching from warming, overfishing (giant triton / crown of thorns starfish), introduced species (red lionfish), loss of mangroves and seagrass nurseries • Conservation: marine protected areas / national parks (Great Barrier Reef, Belize, Chagos), mooring buoys, diver education, No Take Zones which can increase catches elsewhere, sustainable ecotourism and fishing, protection of mangroves and seagrass, erosion and pollution control, planning control of ports and resorts • Fishing controls: closed seasons, minimum and maximum catchable sizes, bans on dynamite, trawling and small-mesh nets • Evaluation: local measures cannot control global threats (warming, acidification) which need international action; enforcement in remote areas is difficult; conflicts with local livelihoods; large vs small NTZs • Justified conclusion	9
	Model answer - Level 3 (9 marks)	

Question	Marking guidance	Marks
	Tropical coral reefs cover about one per cent of the oceans yet support over a quarter of marine species, a diversity built on the symbiosis between polyps and zooxanthellae that recycles nutrients in nutrient-poor water. They matter to people for fisheries of fish, lobsters and conch; for medicines such as AZT, Ara-C, dolastatin, secosteroids, bryostatin and trabectedin discovered in reef organisms; as a carbon sink, since every carbonate unit in coral skeleton absorbs a molecule of carbon dioxide; for dissipating wave energy that would erode low-lying islands; and for tourism income.	
	They are also unusually vulnerable because they need narrow abiotic conditions. Anchors, divers, litter and coral extraction kill polyps; ornaments are collected; sediment from rivers and coastal development smothers them; oil, sewage and nutrients cause algal shading; ocean acidification hinders skeleton formation; warming causes bleaching; overfishing of giant tritons lets crown-of-thorns starfish destroy reefs; introduced lionfish eat native fish; and loss of mangroves and seagrass removes the nurseries and sediment traps reefs depend on.	
	Marine protected areas such as the Great Barrier Reef, Belize and Chagos combine mooring buoys, diver education, no-take zones that raise catches outside them, sustainable tourism and protection of linked habitats, while fishing controls set closed seasons, minimum and maximum sizes and ban dynamite, trawling and small mesh. These work well against local threats where they are enforced, and no-take zones have measurable benefits, but they cannot touch warming and acidification, which need international action, enforcement in remote areas is weak, and restrictions conflict with local livelihoods. Local conservation buys time; only global climate action can secure the reefs.	
	Weaker answer - Level 1 (3 marks)	
	Coral reefs are important because lots of fish live there and people catch them for food. Tourists visit reefs to dive and this brings money. Reefs also protect the coast from waves. Reefs are threatened by climate change which causes bleaching, and by pollution and damage from boats. They are conserved by making marine protected areas where fishing is banned. This helps but climate change is still a problem.	
	What makes the difference	
	• Level 1 has the right headings but Level 3 explains each (why reefs are diverse, why they are a carbon sink, how each threat acts).	
	• Level 3 uses named reserves and specific management measures; Level 1 gives one generic measure.	
	• Level 3's evaluation separates local from global threats and reaches a nuanced judgement; Level 1 states that climate change is 'still a problem'.	
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 57-62; AQA spec 3.1.2.3.4.	

Question	Marking guidance	Marks
11.1	Discuss the methods that can be used to conserve biodiversity and evaluate their effectiveness. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Setting priorities: IUCN Red List criteria, EDGE, endemic, keystone, flagship species, population dispersal • Legislation and protocols: Wildlife and Countryside Act, designated areas (SSSI to MCZ), CITES Appendices, IWC, EU CFP, ITTO - successes and limits of enforcement • Captive breeding and release: selection, difficulties, techniques, release site criteria, hard / soft release, monitoring; seed banks • Habitat conservation: creation, design (area, shape, corridors, age structure, colonisation), abiotic and biotic management, plagioclimax maintenance, population control, reintroductions • Habitat case studies (woodland, rainforest, coral reefs, islands, mangroves, Antarctica) illustrating threats and conservation efforts • Monitoring and new technologies informing decisions • Evaluation: examples of success (Iberian lynx, Hawaiian goose, Scilly rats, Yellowstone) and failure (ITTO, tigers); cost, politics, climate change; judgement on the most effective combination Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 26-71; AQA spec 3.1.2.3.	25
	Model answer - Level 5 (23 marks)	
	Biodiversity is lost through direct exploitation, habitat destruction, introduced species, pollution and changes to abiotic factors, so its conservation needs methods that address each cause, at scales from a single species to international law. Their effectiveness varies with the threat, the species and the resources available.	

Question	Marking guidance	Marks
	Deciding what to conserve comes first. The IUCN Red List uses population size, area, decline rate and probability of extinction to prioritise species, and endemic, keystone, flagship and evolutionarily distinct species receive weight because their loss has wider consequences. This is effective at directing scarce resources, but it favours well-studied vertebrates over invertebrates and plants.	
	Legislation and agreements protect species and habitats. In the UK the Wildlife and Countryside Act, SSSIs, NNRs and Marine Conservation Zones, and in Europe SACs, SPAs and the Habitats Directive, restrict damaging activities; internationally CITES controls trade in threatened species, the IWC regulates whaling, the Ramsar Convention protects wetlands and the Rio Convention on Biological Diversity commits nations to conservation. Their effectiveness depends on enforcement: CITES has reduced trade in ivory and rhino horn but poaching continues where enforcement is weak; the IWC moratorium allowed some whale populations to recover but Japan and Norway continued whaling; designations protect against development but not against pollution or climate change crossing boundaries.	
	In-situ habitat conservation and management are the most fundamental methods because they keep species with the community and abiotic conditions they depend on. Management uses ecological understanding: maintaining plagioclimaxes such as chalk grassland and heathland by grazing or burning, coppicing and pollarding woodland, controlling water levels, light and grazing, designing reserves with the right area, shape, biological corridors and buffer zones, and creating new habitats such as wetlands in worked-out gravel pits. Case studies show it works - coppiced woodland restored dormice and butterflies, wetland creation increased bitterns - but reserves are often small and fragmented and need permanent funding.	
	Ex-situ methods rescue species for which in-situ protection has failed. Captive breeding faces problems of space, diet, breeding triggers, social groups and small gene pools, addressed with studbooks, artificial insemination, embryo transfer and cryopreservation; seed banks such as the Millennium Seed Bank store plant genetic diversity cheaply. The Arabian oryx, California condor and pool frog show captive breeding can bring species back, but only where protected habitat exists for hard or soft release with post-release monitoring, and it is expensive and works for few species.	
	Ecological monitoring - image databases, eDNA, tracking, satellites - measures whether any of this is working and has improved the targeting of effort. Overall, legislation is effective when enforced, habitat management is the most effective and widely applicable method, and captive breeding is a last resort that succeeds only in combination with habitat protection. The Living Planet Index continues to fall, however, which shows that conservation methods are winning individual battles while the drivers of loss - population, consumption, land-use change and climate change - remain unaddressed.	
	What makes the difference	
	The essay is structured by type of method and evaluates the effectiveness of each in turn, with named examples for both successes and failures.	

Question	Marking guidance	Marks
	It draws on several specification areas (biodiversity, conservation legislation, habitat management, captive breeding, ecological processes, monitoring) - the synoptic breadth Level 5 requires.	
	The conclusion ranks the methods and then steps back to the bigger picture (Living Planet Index, underlying drivers) - a supported judgement rather than a summary.	
	A Level 3 essay would describe the same methods accurately but with fewer examples, little evaluation of enforcement or cost, and a conclusion that all methods are important.	
11.2	Discuss the extent to which the resources and services provided by other species justify the conservation of biodiversity. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) - Resources: wood, fibres, oils, fuels, food; new food species (morama bean, yeheb, perennial maize, Kernza, bison, eland) and their adaptations - Biomimetics: wing-tip feathers, whale tubercles, shark skin, termite mounds, bird bones, gecko adhesion, burrs, lotus, spider silk - Medicines: poppies, cinchona, willow, yew (taxol), Mexican yam, sponges (AZT), sharks; few species yet studied; physiological research (marsupials, hippo, sponges) - Genetic resources: crop wild relatives, Vavilov centres, gene pools and inbreeding - Ecosystem services: atmospheric regulation, hydrological cycle, biogeochemical cycles, soil maintenance; interspecies relationships - food, pollination (Darwin's orchid), seed dispersal, habitat provision - Evaluation: economic vs intrinsic / moral arguments; services unnoticed because of dynamic equilibrium; unknown future value; justified conclusion Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 10-19; AQA spec 3.1.2.1.	25